

Research Proposal: Graph-Based Decentralized Learning for Scalable Multi-Robot System Trajectory Optimization

1. Specific Aims

Autonomous multi-robot systems such as distributed rover teams and spacecraft swarms are essential for the future of planetary explorations and large-scale science missions. However, current onboard planning algorithms face a critical trade-off between mathematical rigor and computational scalability. Optimization-based planners (e.g., Sequential Convex Programming) provide safety guarantees but scale poorly with the number of agents and environmental complexity, often exceeding the capabilities of resource-constrained flight hardware. Conversely, recent "amortized optimization" methods like TOAST (Trajectory Optimization with Merit Function Warm Starts) utilize neural networks to warm-start solvers, reducing computation time by orders of magnitude. [1]

However, a fundamental limitation exists in the current state-of-the-art: methods like TOAST rely on fixed-size input vectors to represent the environment (e.g., hard-coding a specific number of obstacles). This "Input Rigidity" renders the architecture permutation-variant and non-scalable; if the number of agents changes or dynamic obstacles appear, the model fails, requiring retraining from scratch.

This research proposes to overcome this limitation by developing a Graph-Based Decentralized Warm Starting Framework. By replacing the fixed-vector architecture with Graph Neural Networks (GNNs), we can learn permutation-invariant policies that scale to arbitrary swarm sizes. The central hypothesis is that Graph Neural Networks can effectively encode variable local neighborhoods to predict optimal dual variables and control strategies, enabling decentralized agents to warm-start their local trajectory optimizers without central coordination or retraining.

Research Aims:

1. Develop a Graph-Based Warm-Start Architecture: We will replace the fixed-input Multilayer Perceptron (MLP) used in TOAST with a permutation-invariant Graph Attention Network (GAT). The multi-robot system will be encoded as a graph $\mathcal{G}=(\mathcal{N}, \mathcal{E})$, enabling the model to process variable swarm sizes without retraining.[2]
 - Node Features (V): Each agent node v_i will encode its local state vector, including position, velocity, fuel mass, and relative goal heading.
 - Edge Features (E): Edges e_{ij} will encode relative kinematics (distance, bearing, relative velocity) and collision risk metrics between agent i and its neighbors $j \in \mathcal{N}_i$.
 - Network Output: Unlike standard Reinforcement Learning (RL) policies that output control actions directly, our GAT will predict the primal-dual initialization

pair $(\hat{x}_0, \hat{\lambda}_0, \hat{\mu}_0)$ estimating both the optimal trajectory and the Lagrange multipliers for active constraints. This allows the network to effectively "warm-start" the onboard Sequential Convex Programming (SCP) solver with a high-quality, feasible guess.

2. Formulate a Decentralized Lagrangian Merit Function: We will extend the constraint-informed loss function of TOAST to a decentralized multi-agent setting. [1, 2]
 - Hybrid Optimization Strategy: The proposed framework utilizes a "Hybrid" control architecture. The GNN provides a rapid, amortized prediction of the optimal solution, which serves as the initialization for the rigorous Sequential Convex Programming (SCvx) solver. This ensures that even if the neural network prediction is imperfect, the downstream optimizer guarantees strict satisfaction of flight-critical safety constraints (e.g., collision avoidance, thrust limits).
 - Distributed Loss: We will formulate a distributed loss function L_{total} that penalizes violations of the Karush-Kuhn-Tucker (KKT) conditions. By training on the "price" of constraints (dual variables λ, μ), the network learns to inherently respect safety margins during the offline training phase, minimizing the correction steps required by the flight solver online.
3. Validate Scalability and Robustness: Demonstrate that the framework trained on small teams (e.g., 3 rovers) generalizes zero-shot to larger swarms (e.g., 10+ rovers) in dynamic lunar analog environments, outperforming fixed-vector baselines.

2. Research Strategy

A. Significance

Future mission concepts, such as the Cooperative Autonomous Distributed Robotics Explorations (CADRE) and long-range endurance rovers, require agents to operate in dynamic, communication-limited environments. [2]

The Bottleneck

Current learning-based trajectory planners are environment-dependent. For example, the TOAST framework trains on a static parameters vector $\theta \in \mathbb{R}^{n_p}$ representing a fixed number of obstacles. This requires training a distinct model for every possible configuration of the environment, which is intractable for real-world deployment. [1]

The Gap

While Multi-Agent Reinforcement Learning (MARL) utilizing Graph Attention Networks (GATs) has shown promise in communication routing (e.g., GAT-MARL for LDTNs), these methods have not yet been bridged with hard-constraint trajectory optimization to guarantee flight-critical safety. [2]

The Impact

Solving this enables permutation-invariant autonomy. A swarm of rovers can lose a member or gain a new one, or encounter a cluster of unexpected obstacles, and the onboard neural network will still function correctly without retraining. This directly supports NASA's goal for robust multi-agent coordination in uncertain Lunar/Martian environments. Beyond space robotics, the framework could inform robust autonomy across air, land, and maritime systems.

B. Innovation

This research introduces a novel class of Graph-Informed Optimization algorithms that merge the structural flexibility of GNNs with the rigorous guarantees of nonlinear programming:

- **Graph-Based State Representation:** Unlike previous work that flattens the environment into a vector, we represent the swarm as a graph $\mathcal{G}=(\mathcal{N}, \mathcal{E})$, where nodes represent agents/obstacles and edges represent relative kinematics. This utilizes attention mechanisms (inspired by GAT-MARL) to dynamically weight the importance of neighbors, allowing the model to handle variable input sizes natively. [3]
- **Decentralized Constraint-Informed Learning:** We innovate on the TOAST "Merit Function" by introducing a Distributed Lagrangian Loss. the network learns to predict not just the primal control variables, (u), but also the dual variables (λ) associated with inter-agent collision constraints. this effectively "teaches" the neural network the "price" of violating safety constraints during the offline training phase. [2]
- **Scalable Generalization:** By training on local subgraph rather than global states, the system allows for zero-shot generalization. A model trained on a 3-rover scenario can be deployed on a 10-rover swarm, significantly reducing the data and training overhead required for mission design. [2]

C. Approach

Aim-1: Graph-Based Architecture for Decentralized Inference

Data Generation

We will generate a dataset of multi-agent trajectory optimization problems using Successive Convexification (SCvx) as the "expert" solver. The dataset will include scenarios with varying numbers of agents (3 to 10) and obstacles to ensure diversity. [4]

Network Design

We will design a Graph Neural Network (GNN) encoder that takes the local observation of an agent i (relative positions/velocities of neighbors $j \in \mathcal{N}_i$) as input.

Mechanism

Utilizing Graph Attention layers (GAT), the network will aggregate features from neighbors to produce a fixed-size embedding regardless of the neighbor count. This embedding will be decoded to predict the warm-start trajectory and dual variables for agent i . [2]

Aim-2: Distributed Lagrangian Merit Function

Loss Formulation

We will extend the Lagrangian loss function $L_{\{\mathcal{L}\}}$ defined in TOAST to the multi-agent setting. [1]

$$L_{\text{total}} = \sum_i (f(\hat{x}_i) + \hat{\lambda}_i^T g_{\text{local}}(\hat{x}_i) + \sum_{j \in \mathcal{N}_i} \hat{\mu}_{ij}^T g_{\text{coll}}(\hat{x}_i, \hat{x}_j))$$

Constraint Encoding

The loss function will explicitly penalize violations of the KKT conditions for collision avoidance. This allows the network to learn a policy that inherently respect safety margins, minimizing the correction steps required by the onboard solver. [1]

Training

We will employ Centralized Training, Decentralized Execution (CTDE). During offline training, the loss is calculated using global state information. During online execution, the agent uses only its local graph to infer the solution. [2]

Aim-3: Validation and Scalability Analysis

Simulation

We will utilize the Multi-Spacecraft Concept and Autonomy Tool (MuSCAT), or GPU powered simulation engines like NVIDIA's Isaac Sim or Python-based equivalent to simulate swarm dynamics. [5]

Benchmarks

We will compare the proposed Graph-Informed SCvx approach against three distinct baselines to quantify performance:

1. Standard TOAST (Fixed-Vector MLP): To demonstrate the "Input Rigidity" failure mode. We expect this baseline to fail or require retraining when the number of agents changes from N to $N+1$. [1]
2. Cold-Start SCvx: Standard optimization without learning. This serves as the baseline for quantifying computational speed-up (targeting a 10x reduction in solve time). [4]
3. GAT-MARL (Pure RL): A pure Reinforcement Learning (RL) approach without a downstream solver. While scalable, Pure RL lacks mathematical safety guarantees. We expect this baseline to exhibit a higher rate of constraint violations (lower Feasibility Rate) during "tail events" or complex obstacle encounters compared to our hybrid approach, highlighting the necessity of the optimization layer for flight certification. [2]

Metrics

We will measure Feasibility Rate (percentage of collision-free trajectories), Solve Time (reduction in ms), and Generalization Gap (performance drop when scaling from N to $2N$ agents)

Technical Challenges and Risk Mitigation

Non-Stationarity in Decentralized Learning

In decentralized setting, the environment becomes non-stationary from the perspective of an individual agent as other agents simultaneously update their policies. This violates standard Markov assumptions. We will mitigate this by conditioning the GNN on the predicted trajectories of neighbors (explicit intent modeling) rather than just their current states, ensuring the policy remains robust to dynamic peer behavior.

Dual Variable Instability

As noted in the original TOAST framework, dual variables (Lagrange multipliers) can exhibit high variance or "explode" during the early phases of training when constraint violations are frequent. We will implement dynamic clipping and log-space normalization strategies to stabilize the Lagrangian value estimation, preventing gradient explosion while preserving the relative "price" of constraints [1]

Data Distribution Shift (Sim-to-Real)

A potential gap exists between the "expert" SCvx trajectories generated offline and the noisy observations encountered online. To address this, we will employ domain randomization (varying obstacle counts and sensor noise) during the data generation phase to ensure the GAT architecture learns a robust representation that generalizes to unseen densities.

Appendix: Prior Research Experience

1. Cognitive Mesh: AI-Driven Resilient Routing for Contested pLEO Architectures

Role: Principal Investigator / Lead Researcher

Summary: Addressed the fragility of deterministic routing tables (e.g., OSPF, Dijkstra) in Proliferated Low Earth Orbit (pLEO) satellite constellations. In contested environments ("DDIL"), static schedules fail when links are jammed or nodes are compromised. This project developed a decentralized routing framework where satellites operate as autonomous agents, dynamically re-routing traffic based on local observations of network health rather than global lookup tables.

Technical Approach:

- **Formulation:** Modeled the pLEO routing problem as a Partially Observable Markov Decision Process (POMDP), where each satellite observes only its immediate neighbors (queue depth, Signal-to-Noise Ratio).
- **Algorithm:** Utilized **Multi-Agent Reinforcement Learning (MARL)** with a **Centralized Training, Decentralized Execution (CTDE)** paradigm. This allowed agents to learn cooperative strategies offline using global state information, while executing policies online using only local data ensuring scalability to 500+ node constellations.
- **Robustness:** Trained agents using Proximal Policy Optimization (PPO) within the Hypatia simulation environment against a "virtual adversary" that injected stochastic link failures and jamming signatures.
- **Outcome:** The learned policies demonstrated a projected 30% improvement in Packet Delivery Ratio (PDR) and a 20% reduction in latency compared to static shortest-path baselines under active interference.

2. TOAST-ROS2: Constraint-Informed Learning for Warm-Starting Trajectory Optimization

Author: Fatih Karatay **Based on Research by:** Briden, Choi, Yun, Linares, and Cauligi (JPL/MIT) [1]

Link: <https://github.com/fatihkaratay/accel-toast>

Summary:

This repository contains a complete implementation and replication of the TOAST (Trajectory Optimization with Merit Function Warm Starts) [1] framework, applied to a 2D mobile robot obstacle avoidance problem. The project demonstrates how machine learning models trained with Lagrangian-based merit functions can learn to predict high-quality, constraint-aware warm starts for nonlinear optimization solvers (IPOPT), enabling real-time planning in ROS2.

Trajectory optimization on resource-constrained flight hardware is often limited by solver convergence times. This project implements a semi-amortized optimization pipeline that:

1. Uses CasADi and IPOPT to solve thousands of random navigation problems offline, collecting full Primal (x, u) and Dual (λ) solutions.
2. Trains a PyTorch Neural Network using a custom Lagrangian Merit Loss function, forcing the model to learn not just the trajectory, but the "price" of constraints (e.g., obstacle proximity).
3. Deploys the trained model as a ROS2 node that provides "Warm Starts" to an online MPC solver, reducing computation time by over 60%.

3. Creating Human Agents in Simulation Environments (CHASE): Imitation Learning for Unmanned Surface Vehicles

Institution: Johns Hopkins University Applied Physics Laboratory

Role: Research Engineer (AI/ML)

Summary:

The CHASE project focused on developing human-like surrogate agents using imitation learning to enable safe and realistic testing of Human–Machine Interaction (HMI) in Modeling and

Simulation (M&S) environments. The core challenge was assessing risk to human safety when evaluating autonomous systems interacting with unpredictable humans. To address this, CHASE introduced an imitation learning toolkit capable of generating probabilistic, human-behavioral models that emulate human decision-making and cognitive variability. The framework combined Imitation from Observation and uncertainty-based reward shaping to train robust agents without the need for live human feedback. CHASE successfully demonstrated the ability to model dynamic human behavior with high fidelity within simulation environments and progressed the technology from TRL 4 to TRL 6. The system now enables autonomy testers to validate system safety, reliability, and performance under realistic human-interaction conditions—without putting human participants at risk.

Key Technical Contributions:

- Developed a probabilistic imitation learning framework combining behavioral cloning and uncertainty-driven rewards.
- Integrated imitation agents within multi-domain simulators.
- Advanced the state of test and evaluation for human–autonomy teaming and system safety analysis.

4. A Rigorous Benchmark of Faithfulness and Robustness in Explainable AI for Medical Vision Transformers

Institution: Johns Hopkins University (EN.705.742 Advanced Applied Machine Learning)

Summary:

This study performed one of the first quantitative evaluations of explainable AI (XAI) techniques for Vision Transformers (ViTs) in computational pathology. Using the PatchCamelyon (PCam) dataset for metastatic cancer detection, three leading XAI methods, Attention Rollout, Integrated Gradients, and ViT-Shapley were benchmarked based on faithfulness (Area Over Perturbation Curve, AOPC) and robustness (Average Sensitivity).

Results demonstrated that Integrated Gradients achieved the most faithful and robust explanations, outperforming ViT-Shapley by over 25× in robustness. This work established a reproducible framework for evaluating transparency in deep models applied to high-stakes medical imaging contexts.

Key Technical Contributions:

- Designed a quantitative benchmarking pipeline for comparing XAI methods.
- Introduced faithfulness and robustness metrics for ViT interpretability.
- Provided reproducible open-source implementations and evaluation scripts.

Impact:

This work supports the development of trustworthy and interpretable deep learning models, a principle that directly extends to explainable autonomy and safety-critical decision systems for robotics.

5. Multi-Sensor Fusion for Target Identification Using Dempster–Shafer Theory

Institution: Johns Hopkins University (EN.605.745 – Reasoning Under Uncertainty)

Summary:

This project explored the application of Dempster–Shafer (DS) evidence theory to enhance target identification and environmental perception in autonomous systems. Using simulated camera, lidar, and radar data, the framework fused multiple uncertain and potentially conflicting sensor inputs to estimate target identities.

Unlike traditional probabilistic fusion, DS theory explicitly models ignorance and evidential uncertainty, producing intervals of belief and plausibility rather than single-point probabilities.

Key Technical Contributions:

- Implemented DS-based evidence fusion using the pyds Python library.
- Designed three benchmark scenarios: high agreement, high conflict, and high ambiguity.
- Quantified conflict coefficients and uncertainty intervals to evaluate fusion robustness.

Impact:

Demonstrated how evidential reasoning improves robustness in multi-sensor autonomous perception, providing a foundation for future work in uncertainty-aware trajectory optimization and sensor-informed autonomy.

References

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